

Advanced Linear Algebra: Foundations to Frontiers

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1.2.5 Unit ball ¶

fit width

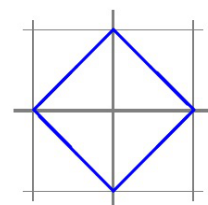
In 3-dimensional space, the notion of the unit ball is intuitive: the set of all points that are a (Euclidean) distance of one from the origin. Vectors have no position and can have more than three components. Still the unit ball for the 2-norm is a straight forward extension to the set of all vectors with length (2-norm) one. More generally, the unit ball for any norm can be defined:

Definition 1.2.5.1. Unit ball. Given norm $\|\cdot\| : \mathbb{C}^m \rightarrow \mathbb{R}$, the unit ball with respect to $\|\cdot\|$ is the set $\{x \mid \|x\| = 1\}$ (the set of all vectors with norm equal to one). We will use $\|x\| = 1$ as shorthand for $\{x \mid \|x\| = 1\}$.

Homework 1.2.5.1. Although vectors have no position, it is convenient to visualize a vector $x \in \mathbb{R}^2$ by the point in the plane to which it extends when rooted at the origin. For example, the vector $x = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ can be so visualized with the point (2, 1). With this in mind, match the pictures on the right corresponding to the sets on the left:

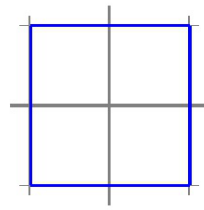
(a) $\|x\|_2 = 1$.

(1)



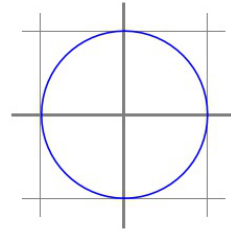
(b) $\|x\|_1 = 1$.

(2)



(c) $\|x\|_\infty = 1$.

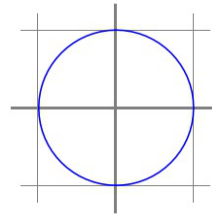
(3)



▼ Solution

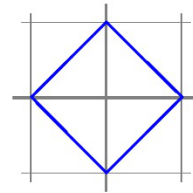
(a) $\|x\|_2 = 1$.

(3)



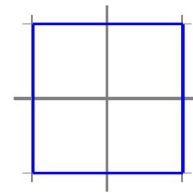
(b) $\|x\|_1 = 1$.

(1)



(c) $\|x\|_\infty = 1$.

(2)



fit width